



Charging infrastructure can be boosted tremendously through the right shift of data

design approach because the two do not sit together and work hand in hand. This issue disappears with virtual simulation as there is now a platform for them to work remotely and yet coordinate with each other, thus enhancing the time to market and quality of the product.

Digitalisation in battery management

Batteries are one of the most important aspects of EV. As such, battery development and management need to be efficient, which is possible through digitalisation. Battery development includes processes such as thermal simulation, battery behaviour, battery life management, and charging.

“There is an opportunity for more democratisation of energy where both the manufacturing side as well as the generation and management of the consumption of energy can be better monitored through digitalisation,” explains Bharath Krishna Rao, Managing Director, Emobi Manufacturing Ltd.

Machine learning (ML) is another critical digital aspect that can be leveraged by the EV industry, especially in terms of battery packs where there is huge field case data. This data can be used to train the process and devise a system through

ML for certain eventualities and the possible responses.

Similarly, tracing and tracking the battery cells all through their life-cycle to assess their behaviour in different climatic conditions and see their performance in different pack configurations, and different form factors for different vehicle level applications, also provides crucial data. Pulling all that data into a common platform, AI/ML algorithms can be used to determine important parameters that set the tone for better battery chemistries to be developed in the future.

This data also helps in integrating other components of the EV, such as controllers and vehicle management systems, to communicate and work along with the battery by exchanging crucial data points. “This will allow us to expand at a larger and faster pace to make more microfactories viable, where the core elements of the vehicle management system can be consolidated by core OEMs, while you can have several assembly partners who could help increase the pace of deployment of EVs into global markets. This is only possible with really integrating with digitalisation right from the suppliers’ facility all the way to the consumer’s hand and back from the consumer for

second-life use case based on the state of the battery or being used for recycling based on the state of the battery,” adds Rao.

Boosting charging infrastructure

With digitalisation comes globalisation as it becomes easy to communicate with different factories around the world through integrated IT platforms run on IoT and cloud. The digitalisation at global level helps not only in the charging infrastructure, it is also used in selecting the right energy supplier and buying and selling energy.

The charging infrastructure, for one, can be boosted tremendously through the right shift of data. For instance, in smart charging, whenever a user plugs in an EV, information related to charging speed and time, the charge-point usage pattern, and the grid capacity can be sent to the cloud and analysed in real time to gain insights to manage the grid capacity.

Innovations like wireless charging, which India is yet to catch up with, also demand use of digital tools to their full potential.

Challenges in digitalisation

It is common among EV OEMs in the two- and three-wheeler seg-